

# 1 April 25th, 2020 Meeting

## 1.1 Our Paper

**Nick's Question 1.1** I think we can strengthen Theorem 3.2 from our paper for when we have a  $L^2_\rho$  initial condition. The result from Theorem 3.2 is that if we start with a SPDE

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2}\Delta_x u(t, x) = b(x, u(t, x))\dot{W}(t, x) & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = \mu(\cdot) \end{cases}$$

and then consider

$$\begin{cases} \frac{\partial v}{\partial t}(t, x) - \frac{1}{2}\Delta_x v(t, x) = b(x, v(t, x))\dot{W}(t, x) & x \in \mathbb{R}^d, t > 0 \\ v(0, x) = u(t_0, x) \end{cases}$$

then this new SPDE satisfies

$$\sup_{t \geq 0} \mathbb{E}|v(t, \cdot)|_\rho^2 < \infty.$$

But  $v(t, x) = u(t + t_0, x)$  so really what this theorem is saying is that

$$\sup_{t \geq t_0} \mathbb{E}|u(t, \cdot)|_\rho^2 < \infty.$$

But I was thinking about what happens for  $t \in [0, t_0]$ . From your paper with Huang [15], you proved that

$$(t, x) \mapsto E(|u(t, x)|^2)$$

is continuous on  $(0, \infty) \times \mathbb{R}^d$  if assuming the appropriate conditions listed in [15]. But I think that this can imply that

$$t \mapsto E(|u(t, \cdot)|_\rho^2)$$

is continuous on  $(0, \infty)$  and this is sort of what Theorem 2.3 from the Tessitore paper is saying [40].

Now, are there situations where we can say that

$$t \mapsto E(|u(t, \cdot)|_\rho^2)$$

is continuous on  $[0, \infty)$ ? This can't be the case if we have  $\delta_0$  as the initial condition because

$$E(|u(0, \cdot)|_\rho^2) = E(|\delta_0(\cdot)|_\rho^2)$$

does not exist. But I believe that if we have a  $\varphi \in L^2_\rho$  initial condition, then

$$t \mapsto E(|u(t, \cdot)|_\rho^2)$$

is continuous on  $[0, \infty)$ . More specifically, this would imply that

$$\sup_{0 \leq t \leq t_0} \mathbb{E}|u(t, \cdot)|_\rho^2 < \infty$$

which further implies that

$$\sup_{t \geq 0} \mathbb{E}|u(t, \cdot)|_\rho^2 < \infty.$$

This would be an improvement to the result we already proved.

## 1.2 Misiat and Stanzhytski Paper

**Remark 1.1.** A somewhat stonger result has already been proven in Misiats and Stanzhytski paper [30]. In their paper they consider the SPDE

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - Au(t, x) = f(x, u(t, x)) + b(x, u(t, x))\dot{W}(t, x) & x \in G, t > 0 \\ u(0, \cdot) = \mu(\cdot) \end{cases}$$

where  $A$  is any elliptic operator and  $G \subseteq \mathbb{R}^d$ . They proved that when starting with a  $L^2$  function, hence not any  $L^2_\rho$  function, then

$$\sup_{t \geq 0} \mathbb{E}|u(t, \cdot)|_\rho^2 < \infty.$$

However, in their paper they require that there exists a function  $\psi \in L^1 \cap L^\infty$  and constant  $\sigma_0 > 0$  such that

1.  $|b(x, u)| \leq \sigma_0$
2.  $|f(x, u)| \leq \psi(x)$  for any  $(x, u) \in \mathbb{R}^d \times \mathbb{R}$

I don't believe that we required the first assumption to get the result but we only considered  $f \equiv 0$ , so if we do not assume that then I am not sure if we will also need the second assumption or not. However our proof relies on the Moment estimates from [15] which only apply to the case of  $A = \frac{1}{2}\Delta$ . Also the moment bounds are for the case for  $f \equiv 0$  so extra work is going to need to be done if we are going to want to relax the second condition above which is used in [30]. However, since

$$\int_0^t \int_{\mathbb{R}^d} f(s, u(s, y))G(t-s, x-y) \, dx dy$$

is not a stochastic integral then we could copy the proof in [30] which deal with this integral exactly and keep the second condition listed above. This may help us to prove the existence of an invariant with non-zero  $f$  term with the  $\delta_0$  initial condition.

**Nick's Question 1.2** In Theorem 1 in the paper [30]. They are trying to show that

$$E \left| \int_0^t \int_{\mathbb{R}^d} G(t-s, x-y) \sigma(y, u(s, y)) dy dW(s, y) \right| < \text{Const.}$$

Here is a part of what they did:

$$\int_0^{t-1} \int_{\mathbb{R}^d} G^2(t-s, y) \, dy ds \leq C_1 \int_0^{t-1} \frac{1}{(t-s)^{d/2}} \, ds \leq C_2 < \infty$$

I don't see why the first inequality is true unless we are dealing with heat equation for which it would then be true since  $G(t, x)$  would be the standard heat kernel

$$\begin{aligned} \int_0^{t-1} \int_{\mathbb{R}^d} G^2(t-s, y) \, dy ds &= \int_0^{t-1} G(2(t-s), 0) \, ds \\ &\leq \int_0^{t-1} \frac{1}{(t-s)^{d/2}} \, ds \end{aligned}$$

Also Misiat and Stanzhytski never make mention of any restriction on the correlation function of the Gaussian noise. What they are essentially doing is finding a bound on the second moment but for this to be done, we need to make assumptions on the correlation

function of the Gaussian noise, namely we need to assume

$$\int_{\mathbb{R}^d} \frac{f(d\xi)}{|\xi|^2} < \infty$$

**Remark 1.2 (Uniqueness of the invariant measure).** There is a property called exponential stability defined in [30]: A solution of an SPDE  $u$  is said to be exponentially stable if there exists  $K > 0$  and  $\gamma > 0$  such that for any  $t_0$  and any other solution  $\eta$  with  $\eta(t_0) \in \mathcal{F}_{t_0}$  and  $\mathbb{E}\|\eta(t_0)\|_H^2 < \infty$ , we have that

$$\mathbb{E}\|u(t) - \eta(t)\|_H^2 \leq K \exp(-\gamma(t - t_0)) \mathbb{E}\|u(t_0) - \eta(t_0)\|_H^2$$

for any  $t \geq t_0$ .

This property can be used to show the existence and uniqueness of an invariant measure. In [30], they show that the solution to

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - Au(t, x) = f(x, u(t, x)) + b(x, u(t, x))\dot{W}(t, x) & x \in G, t > 0 \\ u(0, \cdot) = u_0(\cdot) \end{cases}$$

where

- $G$  is the upper half plane
- $\rho(x) = \exp(-|x|^2)$
- $Au := \frac{1}{\rho} \operatorname{div}(\rho \nabla u)$
- $u_0 \in L_\rho^2$
- Both  $f$  and  $b$  are Lipschitz in the second variable with LIP constant independent of  $x$  and  $f(x, 0)$  and  $b(x, 0)$  are in  $L^\infty$

has this exponential stability property and then refer to two propositions (11.2 and 11.4) in [20] which give the existence and uniqueness of an invariant measure.

In general, if we are given SPDE

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - Au(t, x) = f(x, u(t, x)) + b(x, u(t, x))\dot{W}(t, x) & x \in G, t > t_0 \\ u(0, \cdot) = u_0(\cdot) \end{cases}$$

with solution  $u(t, t_0, u_0)$  that satisfies the exponential stability property and in addition satisfies the semigroup property (like the solution to the heat equation does), then we can show that for any  $t_0$ , that  $u(t, t_0, u_{t_0}) \rightarrow \mu$  as  $t \rightarrow \infty$  where  $\mu$  is invariant and  $\mu = \mathcal{L}(u(t_0))$ .

### 1.3 Case when $f \neq 0$

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2} \Delta_x u(t, x) = f(x, u(t, x)) + b(x, u(t, x))\dot{W}(t, x) & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = \mu(\cdot) \end{cases}$$

[2] proves the exist of an invariant measure when the initial condition is a function  $\varphi_0(\cdot) \in L_\rho^p(\mathbb{R}^d)$  where the  $p = 2 \wedge \nu$  and  $\nu$  depends on the function  $f$ . I have not read this paper yet but I plan on it so I am not sure how they prove this.

I have tried to see if the proof from our paper (which is a mimic of the proof by Tessitore in [40]) would work but I ran into the following issue. Our proof for the existence of an invariant

measure relies on the the fact that for fixed  $0 < \delta_0 \leq 1$  we can find in this set  $\mathcal{K}(\Theta) \subset L_\rho^2$  defined in the following way

$$\mathcal{K}(\Theta) = \{(G(\delta_0, \cdot) * y(\cdot))(x) + (F_\alpha^{\delta_0} h)(x), \quad \text{where } |y|_{\hat{\rho}} \leq \Theta \text{ and } |h|_{L^q(0,1,L_\rho^2)} \leq \Theta\}$$

and if we still assume that  $f \equiv 0$  then we can use the factorization theorem and say that the solution  $u$  can be written as

$$u(t_0, x) = (G(t_0, \cdot) * \varphi(\cdot))(x) + \frac{\sin(\alpha\pi)}{\pi} F_\alpha^{t_0}(Y(s, \cdot))(x)$$

and because of the compactness of the operators  $G$  and  $F_\alpha^{t_0}$ , the set  $\mathcal{K}(\Theta)$  is relatively compact.

The issue when we now consider  $f \not\equiv 0$  is the following. By the factorization theorem, we could now represent  $u$  as

$$u(t_0, x) = (G(t_0, \cdot) * \varphi(\cdot))(x) + \left( G(\circ, \cdot) \star f(\cdot, u(\circ, \cdot)) \right)(t_0, x) + \frac{\sin(\alpha\pi)}{\pi} F_\alpha^{t_0}(Y(s, \cdot))(x)$$

and we could try to mimic our proof and define a new  $\mathcal{K}^*(\Theta)$  as follows

$$\mathcal{K}^*(\Theta) = \left\{ (G(\delta_0, \cdot) * y(\cdot))(x) + \left( G(\circ, \cdot) \star f(\cdot, q(\circ, \cdot)) \right)(\delta_0, x) + (F_\alpha^{\delta_0} h)(x), \right. \\ \left. \text{where } |y|_{\hat{\rho}} \leq \Theta \text{ and } |h|_{L^q(0,1,L_\rho^2)} \leq \Theta \text{ and } q \text{ has some property} \right\}$$

We need this new set  $\mathcal{K}^*(\Theta)$  to be relatively compact in  $L_\rho^2$  but then we would need the operator  $\left( G(\circ, \cdot) \star f(\cdot, q(\circ, \cdot)) \right)(\delta_0, x)$  to be compact so that  $\mathcal{K}^*(\Theta)$  remains relatively compact. So we need to see what space this function  $q$  needs to be in. Note that we want  $f(x, u(t, x))$  to be a specific example of  $f(x, q(t, x))$ . Maybe we want the following

1.  $t \mapsto q(x, t)$  is continuous or even just bounded on  $t \in [0, T]$  for any  $T > 0$ .
2.  $x \mapsto q(x, t)$  is in  $L_\rho^2$

and the solution  $u$  satisfies these properties. However I was unable to show that the operator  $\left( G(\circ, \cdot) \star f(\cdot, q(\circ, \cdot)) \right)(t_0, x)$  was bounded (continuous). Following the steps in [40] we would first want to show something like

$$\left| \left( G(\circ, \cdot) \star f(\cdot, q(\circ, \cdot)) \right)(\delta_0, x) \right|_\rho^2 \leq K |q(\delta_0, \cdot)|_\rho^2$$

However the best I could do was show that

$$\left| \left( G(\circ, \cdot) \star f(\cdot, q(\circ, \cdot)) \right)(\delta_0, x) \right|_\rho^2 \leq K(T) |q(t^*, \cdot)|_\rho^2$$

where

$$q(t, x) \leq q(t^*, x) \quad \text{where we need to assume that } q(\cdot, x) \text{ has a max at } t^* \text{ on the interval } [0, T]$$

Things seem off about this so I will read Assing and Manthey's paper [2] to see how they consider the case when  $f \not\equiv 0$ .

## References

- [1] Amir, Gideon, Ivan Corwin and Jeremy Quastel. Probability distribution of the free energy of the continuum directed random polymer in  $1 + 1$  dimensions. *Comm. Pure Appl. Math.* 64 (2011), no. 4, 466–537.
- [2] Assing, Sigurd and Ralf Manthey. Invariant measures for stochastic heat equations with unbounded coefficients. *Stochastic Process. Appl.* 103 (2003), no. 2, 237–256.
- [3] Balan, Raluca M. and Le Chen. Parabolic Anderson model with space-time homogeneous Gaussian noise and rough initial condition *J. Theoret. Probab.*, to appear 2017.
- [4] V. Bally and E. Pardoux. Malliavin calculus for white noise driven parabolic SPDEs. *Potential Anal.* 9 (1998), no. 1, 27–64.
- [5] Bertini, Lorenzo and Giambattista Giacomin. Stochastic Burgers and KPZ equations from particle systems. *Comm. Math. Phys.* 183 (1997), no. 3, 571–607.
- [6] Bertini, Lorenzo and Giambattista Giacomin. On the long-time behavior of the stochastic heat equation. *Probab. Theory Related Fields*, 114 (1999), no. 3, 279–289.
- [7] Bouleau, Nicolas and Francis Hirsch. *Dirichlet forms and analysis on Wiener space*. De Gruyter Studies in Mathematics, 14. Walter de Gruyter & Co., Berlin, 1991. x+325 pp.
- [8] Cannizzaro, Giuseppe, Peter K. Friz and Paul Gassiat. Malliavin Calculus for regularity structures: the case of gPAM. *J. Funct. Anal.*, 272 (2017), no. 1, 363–419.
- [9] Carmona, René A. and Stanislav A. Molchanov. Parabolic Anderson problem and intermittency. *Mem. Amer. Math. Soc.*, 108 (1994), no. 518, viii+125 pp.
- [10] Cerrai, Sandra. *Second order PDE's in finite and infinite dimension*. A probabilistic approach. Lecture Notes in Mathematics, 1762. Springer-Verlag, Berlin, 2001. x+330 pp.
- [11] Cerrai, Sandra. Stochastic reaction-diffusion systems with multiplicative noise and non-Lipschitz reaction term. *Probab. Theory Related Fields* 125 (2003), no. 2, 271–304.
- [12] Cerrai, Sandra. Asymptotic behavior of systems of stochastic partial differential equations with multiplicative noise. *Stochastic partial differential equations and applications–VII*, 61–75, Lect. Notes Pure Appl. Math., 245, Chapman & Hall/CRC, Boca Raton, FL, 2006.
- [13] Chen, Le and Robert C. Dalang. Moments and growth indices for nonlinear stochastic heat equation with rough initial conditions. *Ann. Probab.* 43 (2015), no. 6, 3006–3051.
- [14] Chen, Le, Yaozhong Hu and David Nualart. Regularity and strict positivity of densities for the nonlinear stochastic heat equation. *Mem. Amer. Math. Soc.*, 2019, to appear.
- [15] Chen, Le and Jingyu Huang. Comparison principle for stochastic heat equation on  $\mathbb{R}^d$ . *Ann. Probab.* 2019, to appear.
- [16] Chen, Le and Kunwoo Kim. Nonlinear stochastic heat equation driven by spatially colored noise: moments and intermittency. *Acta Math. Sci. Ser. B*, 2019, to appear.
- [17] Dalang, Robert C. Extending the martingale measure stochastic integral with applications to spatially homogeneous s.p.d.e.'s. *Electron. J. Probab.* 4 (1999), no. 6, 29 pp.
- [18] Dalang, Robert, Davar Khoshnevisan, Carl Mueller, David Nualart, and Yimin Xiao *A minicourse on stochastic partial differential equations*. Held at the University of Utah, Salt Lake City, UT, May 819, 2006. Edited by Khoshnevisan and Firas Rassoul-Agha. Lecture Notes in Mathematics, 1962. Springer-Verlag, Berlin, 2009. xii+216 pp.
- [19] Dalang, Robert C. and Lluís Quer-Sardanyons. Stochastic integrals for spde's: a comparison. *Expo. Math.* 29 (2011), no. 1, 67–109.

- [20] Da Prato, Giuseppe and Jerzy Zabczyk. *Stochastic equations in infinite dimensions*. Encyclopedia of Mathematics and its Applications, 44. Cambridge University Press, Cambridge, 1992. xviii+454 pp.
- [21] Da Prato, Giuseppe and Jerzy Zabczyk. *Ergodicity for infinite-dimensional systems*. London Mathematical Society Lecture Note Series, 229. Cambridge University Press, Cambridge, 1996.
- [22] Eckmann, Jean-Pierre and Martin Hairer. Invariant measures for stochastic partial differential equations in unbounded domains. *Nonlinearity* 14 (2001), no. 1, 133–151.
- [23] Foondun, Mohammud and Davar Khoshnevisan. On the stochastic heat equation with spatially-colored random forcing. *Trans. Amer. Math. Soc.* 365 (2013), no. 1, 409–458.
- [24] Grafakos, Loukas *Modern Fourier analysis*. Third edition. Graduate Texts in Mathematics, 250. Springer, New York, 2014. xvi+624 pp.
- [25] Goldys, Benjamin, and Bohdan Maslowski. Uniform exponential ergodicity of stochastic dissipative systems. *Czechoslovak Math. J.* 51(126) (2001), no. 4, 745–762.
- [26] Hairer, Martin. A theory of regularity structures. *Invent. Math.*, 198 (2) (2014), 1–236.
- [27] Hairer, Martin. Solving the KPZ equation. *Ann. of Math.*, (2) 178 (2013), no. 2, 559–664.
- [28] Y. Hu, J. Huang, D. Nualart, X. Sun. Smoothness of the joint density for spatially homogeneous SPDEs. *J. Math. Soc. Japan*, Vol. 67, No. 4 (2015), 1605–1630.
- [29] Maslowski, Bohdan and Jan Seidler. On sequentially weakly Feller solutions to SPDE's. *Atti Accad. Naz. Lincei Cl. Sci. Fis. Mat. Natur. Rend. Lincei (9) Mat. Appl.* 10 (1999), no. 2, 69–78.
- [30] Misiats, Oleksandr, Oleksandr Stanzhytskyi, and Nung-Kwan Yip. Existence and uniqueness of invariant measures for stochastic reaction-diffusion equations in unbounded domains. *J. Theoret. Probab.* 29 (2016), no. 3, 996–1026.
- [31] Mueller, Carl and David Nualart. Regularity of the density for the stochastic heat equation. *Electron. J. Probab.*, 13 (2008), no. 74, 2248–2258.
- [32] D. Nualart. *Malliavin calculus and its applications*. CBMS Regional Conference Series in Mathematics, 110. Published for the Conference Board of the Mathematical Sciences, Washington, DC; by the American Mathematical Society, Providence, RI, 2009.
- [33] D. Nualart and L. Quer-Sardanyons. Existence and smoothness of the density for spatially homogeneous SPDEs. *Potential Anal.* 27 (2007), no. 3, 281–299.
- [34] E. Nualart. On the density of systems of non-linear spatially homogeneous SPDEs. *Stochastics*, 85 (2013), no. 1, 48–70.
- [35] F. W. J. Olver, D. W. Lozier, R. F. Boisvert, and C. W. Clark, editors. *NIST handbook of mathematical functions*. U.S. Department of Commerce National Institute of Standards and Technology, Washington, DC. Cambridge Univ. Press, Cambridge, 2010.
- [36] Pardoux, Etienne and Tusheng Zhang. Absolute continuity of the law of the solution of a parabolic SPDE. *J. Funct. Anal.* 112 (1993), no. 2, 447–458.
- [37] Peszat, and Zabczyk Stochastic Evolution Equations with a Spatially Homogenous Wiener Processes. *SNS* (1995)
- [38] Quer-Sardanyons, Lluís and Marta Sanz-Solé. A stochastic wave equation in dimension 3: smoothness of the law. *Bernoulli*, 10 (2004), no. 1, 165–186.
- [39] Tessitore, Gianmario and Jerzy Zabczyk. Strict positivity for stochastic heat equations. *Stochastic Process. Appl.* 77 (1998), no. 1, 83–98.
- [40] Tessitore, Gianmario and Jerzy Zabczyk. Invariant measures for stochastic heat equations. *Probab. Math. Statist.* 18 (1998), no. 2, Acta Univ. Wratislav. No. 2111, 271–287.

- [41] Walsh, John B. *An Introduction to Stochastic Partial Differential Equations*. In: École d'été de probabilités de Saint-Flour, XIV—1984, 265–439. Lecture Notes in Math. 1180, Springer, Berlin, 1986.